

Voicing Your Emotion: Integrating Emotion and Identity in Cross-Modal 3D Facial Animations

Wenfeng Song
Computer School, Beijing
Information Science and
Technology University,
Beijing, China

Zhenyu Lv
Computer School, Beijing
Information Science and
Technology University,
Beijing, China

Xuan Wang
Computer School, Beijing
Information Science and
Technology University,
Beijing, China

Xia Hou*
Computer School, Beijing
Information Science and
Technology University,
Beijing, China

ABSTRACT

Recent advances in speech-driven 3D facial animation have yielded promising results, yet capturing intricate expressiveness, particularly in emotion and identity, remains an intricate challenge. Addressing this deficiency, we unveil a pioneering method tailored to produce 3D facial expressions that resonate deeply with emotion and identity, guided by speech and user-provided prompt words. Our key insight is an emotion-identity fusion mechanism, a pre-trained self-reconstruction codebook, meticulously crafted from a wide array of emotional facial movements, serving as an expressive motion benchmark. Using this foundation, prompt words are seamlessly transformed into facial representations that capture both emotion and identity, and are then adeptly projected onto 3D templates. Our approach emerges as a robust tool for crafting 3D talking avatars, rich in emotional depth and distinctive identity.

Index Terms: Computing methodologies—Multimodal—Cross-modal interaction and perception

1 INTRODUCTION

Dynamic and realistic speech-driven facial animation has been widely used in VR-related applications, such as virtual socializing and 3D animation creation. Scholarly undertakings, have focused on achieving remarkable strides in realizing photorealistic animations and achieving precise lip-synchronization [4]. However, a discernible lacuna persists – the integration of emotional expressiveness and distinct identity, elements that transform technical renderings into lifelike avatars. In this study, to amalgamate diverse speech styles of characters and create a diverse tapestry of expressions mirroring the dynamic nature of the human experience, we introduce an emotion and identity feature fusion module rooting in the premise of integrative expressiveness. Furthermore, to bridge the gap between textual prompts and 3D physical states, we propose a cross-modality 3D talking face generation method. We offer a customized, flexible, and adaptive solution that resonates with the complexity of real-world scenarios.

2 CROSS-MODAL 3D FACIAL ANIMATIONS WITH EMOTION AND IDENTITY

Although previous works are able to produce accurate lip movements, the character's exterior style is rigid and cannot be customized. Therefore, we expect to obtain facial animations with customizable appearance styles. The specific pipeline of our network is shown in Fig. 1. The entire network consists of four modules. Firstly, pre-trained Wav2Vec2.0 [1] model are used to encode audio to obtain voice signals. Secondly, the Text2Avatar Module receives customized prompts for the appearance of the speaker, thereby generating the desired facial template. Thirdly, the Emotion-Identity Fu-

sion Module embeds the specified expression into facial movements. Finally, the Avatar Generator uses the encoded signal from the previous modules to generate facial animation through an auto-regressive process, querying facial motion priors within the constrained space of the codebook.

2.1 Text2Avatar

To create customized character appearances, we design a Text2Avatar Module. This module takes character prompts as input and generates custom facial templates. These templates are then used by the Avatar Generator module to produce facial speech animations with different emotions. The input for this process consists of several text keywords, and the output is a static human head model.

The workflow of the Text2Avatar module begins by inputting the prompt words. Firstly, we use a diffusion model to generate a facial image. This defines a Markov chain of diffusion steps, slowly adding random noise to the real data, called a forward process. Then, we learn the reverse diffusion process, called a reverse process, which constructs the image data we need from the noise using conditional terms. This image is fed into the facial reconstruction model EMOCA [2], producing the head template. From the reconstructed head template, we select 5023 vertices based on the FLAME [3] model to create a simplified model, resulting in the best performance of the generated facial model while preserving maximum detail.

2.2 Emotion-Identity Fusion Module

The emotion identity fusion module achieves facial expression control and can learn character language style clues. In the module, in order to better embed emotion and identity vectors into facial motions, Stylization Block [5] is applied after Linear Self attention, which brings timestamp t to the generation process. Specifically, two features identity i and emotion e are fused with Linear Self-attention, outputs pass through Stylization Block and are added to the information. The embedding block combines the past facial motions $M_{1:t-1}$ and the style embedding to obtain encoded motion feature via:

$$\mathbf{F}^{1:t-1} = L(\hat{\mathbf{M}}_{1:t-1}) + S(\text{Attn}(\mathbf{B} \cdot \frac{\mathbf{i}}{\|\mathbf{i}\|_1}, \mathbf{B} \cdot \frac{\mathbf{e}}{\|\mathbf{e}\|_1}), \mathbf{B} \cdot \frac{\mathbf{e}}{\|\mathbf{e}\|_1}), \quad (1)$$

where $\mathbf{F}^{1:t-1}$ represents the encoded motion feature, $L(\cdot)$ is the projection layer, $\text{Attn}(\cdot)$ is the Linear Self-attention, $S(\cdot)$ is the Stylization Block, $\mathbf{B}$ represents the adaptable fundamental vectors that form a linear span across the emotion and identity domain, $\mathbf{e}$ represents the conditioned emotions, $\mathbf{i}$ represents the conditioned identity.

2.3 Avatar Generator

We develop an Avatar Generator based on the attention mechanism to generate talking avatars with different appearances, emotions, and corresponding speech. Firstly, we employ a self-supervised pretrained speech model Wav2Vec 2.0 as a primary component to encode the speech. This model comprises an audio feature extractor and a multi-layer transformer encoder. The audio feature extractor transforms raw waveform speech into feature vectors using a

*E-mail: houxia@bistu.edu.cn

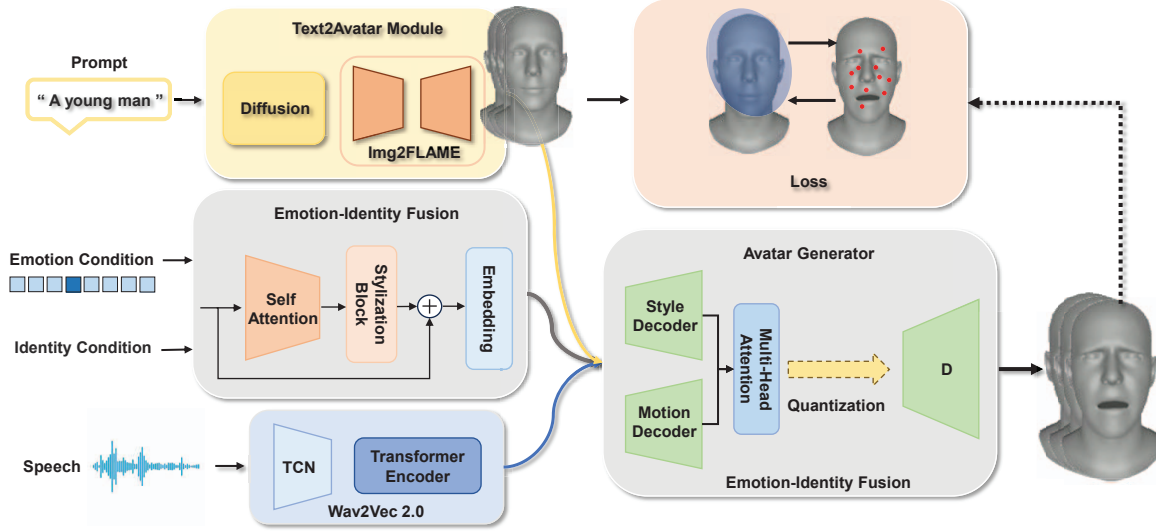


Figure 1: The framework of our method. Input speech audio, specified facial expression types, and prompt words for customized facial templates, and then generate a 3D talking facial model.

temporal convolutional network (TCN). Subsequently, the transformer encoder contextualizes these feature vectors, resulting in contextualized speech features.

Next, we use two transformer decoders equipped with causal self-attention to learn the correlation between each frame in the context of the facial motion sequences. Specifically, the Style Decoder focuses on decoding style and emotional features, while the Motion Decoder focuses on facial motion details. The outputs of the two decoders are fused together with multi-head attention, aligning the audio and motion modalities and preserving rich facial expressions.

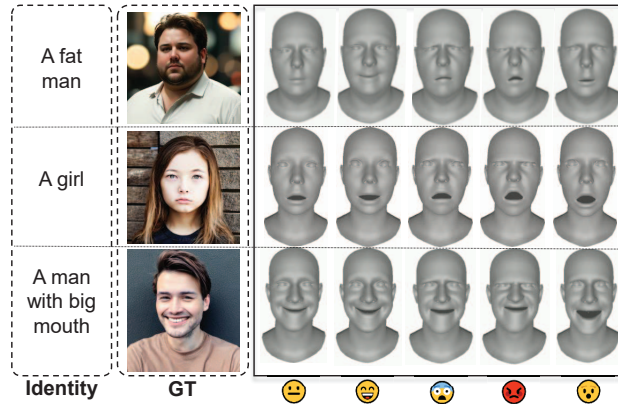


Figure 2: Cross modal generation of talking faces with different emotions for specific identities from prompt words.

Facial motion is a complex phenomenon, as a single lip shape can correspond to multiple facial expressions, and a single speech can be accompanied by various potential facial expressions. It is advantageous to map high-dimensional complex faces into low-dimensional spaces to simplify the mapping process and alleviate ambiguity (especially in one-to-many cases). Additionally, it is important to note that facial movements are interconnected and not independent. Consequently, our approach involves training the VQ-VAE codebook to learn motion sequences as a preliminary step.

The fused output of Style Decoder and Motion Decoder is further quantified and decoded by pre-trained VQ-VAE to obtain the final result. The results are shown in Fig. 2

3 CONCLUSION

We introduced a new method that uses both speech and user-provided words to create 3D faces that not only talk but also express feelings and have their own identity. We used advanced techniques to make sure every animation tells a story, not just through words spoken but also through the emotions expressed and the uniqueness of the character. In simpler terms, our work is about giving life to 3D animations. We have moved beyond just lip movement and speech to create animations that feel real, have their own identity, and can express emotions, making each one special and relatable. This is a big step forward in the world of 3D facial animations.

REFERENCES

- [1] A. Baevski, H. Zhou, A. Mohamed, and M. Auli. wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. *arXiv e-prints*, p. arXiv:2006.11477, June 2020. doi: 10.48550/arXiv.2006.11477
- [2] R. Danecek, M. J. Black, and T. Bolkart. EMOCA: Emotion driven monocular face capture and animation. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 20311–20322, 2022.
- [3] T. Li, T. Bolkart, M. J. Black, H. Li, and J. Romero. Learning a model of facial shape and expression from 4D scans. *ACM Transactions on Graphics, (Proc. SIGGRAPH Asia)*, 36(6), 2017.
- [4] J. Xing, M. Xia, Y. Zhang, X. Cun, J. Wang, and T.-T. Wong. Codetalker: Speech-driven 3d facial animation with discrete motion prior. *arXiv preprint arXiv:2301.02379*, 2023.
- [5] M. Zhang, Z. Cai, L. Pan, F. Hong, X. Guo, L. Yang, and Z. Liu. Motiondiffuse: Text-driven human motion generation with diffusion model. *arXiv preprint arXiv:2208.15001*, 2022.